

JACKSON ESHBAUGH

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Bethlehem, PA, USA

RESEARCH INTERESTS

Computational semantics; computational linguistics; natural language processing; idioms and multiword expressions; neural network interpretability

EDUCATION

- **Lafayette College** August 2023–Expected May 2027
B.S., Computer Science; A.B., French Easton, PA
 - GPA: 4.00/4.00
 - Minor: Mathematics

HONORS AND FELLOWSHIPS

- **Beinecke Scholar** 2026
- **Phi Beta Kappa** Inducted 2026
Lafayette College
- **President, Upsilon Pi Epsilon** Inducted 2026; President, 2026–Present
Lafayette College
- **EXCEL Scholar** May 2025–Present
Lafayette College Department of Computer Science
- **Whitman Fellow** January 2025
Lafayette College Department of Economics
- **Marquis Scholarship** May 2023
Lafayette College
- **Dean's List** All Semesters
Lafayette College

RESEARCH EXPERIENCE

- **Honors Thesis Research—Computational Linguistics** February 2025–Present
Department of Computer Science, Lafayette College Easton, PA
 - Advisor: Dr. Sofia Serrano
 - Formalizing linguistic theories of idiomaticity into computational frameworks for multilingual NLP.
 - Operationalizing dimensions such as semantic opacity, syntactic flexibility, and conventionality for model design and evaluation.
 - Investigating how language models represent idiomatic meaning through interpretability techniques.
- **Independent Study—Linguistics of Idioms** January 2026–May 2026
Department of Language and Literary Studies, Lafayette College Easton, PA
 - Advisor: Dr. Maria Hernandez
 - Examined foundational and contemporary theories of idiomatic language, including structural, construction-based, and cognitive approaches.
 - Analyzed the limitations of structural linguistic models in explaining idiomatic meaning and behavior.
 - Authored a French-language research paper examining the implications of modern linguistic theory for future computational models of idiomaticity.

• **Fully Independent Research—Neural Network Interpretability**

April 2025–Present

Department of Computer Science, Lafayette College

Easton, PA

- Investigated when linear surrogate models fail to faithfully represent neural networks.
- Designed an experimental framework comparing surrogate fidelity and predictive accuracy across regression tasks.
- Demonstrated that high surrogate fidelity does not necessarily imply predictive accuracy and introduced diagnostic measures to characterize this discrepancy.
- Implemented end-to-end machine learning pipelines for neural network training, surrogate evaluation, and statistical analysis.
- Resulted in an accepted workshop paper at ICLR 2026 and an expanded conference manuscript currently in preparation.

• **EXCEL Scholar—Applied Machine Learning for Energy Systems**

May 2025–Present

Department of Computer Science, Lafayette College

Easton, PA

- Advisor: Dr. Jorge Silveyra
- Conduct research on machine learning methods for residential building energy modeling, retrofit analysis, and neighborhood-scale energy planning.
- Develop multimodal generative pipelines and EnergyPlus simulation workflows for synthetic residential building energy datasets.
- First author of a submission on synthetic residential building energy generation (arXiv:2509.09794).
- Presented work at KINBERCON 2025, Lafayette College EXCEL Scholars Poster Sessions, and Georgia State University.
- Preparing manuscripts on calibration-first supervision without ground truth and graph-based algorithms for residential retrofit recommendation.

• **Whitman Fellow—Policy-Oriented Machine Learning**

January 2025

Departments of Economics & Computer Science, Lafayette College

Easton, PA

- Advisors: Dr. Gladstone Hutchinson and Dr. Frank Xia
- Contributed to development of a structured county-level dataset for policy-oriented machine learning research.
- Collected, cleaned, normalized, and integrated heterogeneous public datasets into a shared research database.
- Supported downstream modeling efforts analyzing regional socioeconomic indicators.

PUBLICATIONS

- [1] **Eshbaugh, J.** “Faithful to What?” On the Limits of Fidelity-Based Explanations. 2026. *Workshop on Scientific Methods for Understanding Deep Learning at the International Conference on Learning Representations*. arXiv:2506.12176.
- [2] **Eshbaugh, J.**, Tiwari, C., and Silveyra, J. **A Modular and Multimodal Generative AI Framework for Urban Building Energy Data: Generating Synthetic Homes**. 2026. Preprint. Under review at *Machine Learning with Applications*. arXiv:2509.09794.

PRESENTATIONS

- [1] **Eshbaugh, J.** (Apr 2026). “Faithful to What? On the Limits of Fidelity-Based Explanations.” *Workshop on Scientific Methods for Understanding Deep Learning, International Conference on Learning Representations 2026*, Rio de Janeiro, Brazil. [[poster](#)]
- [2] Silveyra, J. and **Eshbaugh, J.** (Mar 2026). “From Images to Energy Models: Generative AI for Urban Energy Data.” Georgia State University, Atlanta, GA, USA. [[invited talk](#); [slides](#)]
- [3] **Eshbaugh, J.** (Dec 2025). “Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data.” *Lafayette College EXCEL Scholars Poster Session*, Easton, PA. [[poster](#)]

- [4] **Eshbaugh, J.** (Oct 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *CyberAccelerate Poster Session at KINBERCON 2025*, Lancaster, PA. [poster]
- [5] **Eshbaugh, J.** (Sep 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *Lafayette College Bicentennial Weekend Poster Session*, Easton, PA. [poster]

TEACHING EXPERIENCE

- **Teaching Assistant** August 2024–Present
Department of Computer Science, Lafayette College Easton, PA
 - Support instruction during class meetings by assisting students with programming, debugging, and conceptual problem solving.
 - Provide individualized help with course material in introductory computer science contexts.
 - Recognized by students for clear explanations, accessibility, and care for student learning.
- **Mentored Study Group Leader Supporting Computer Science** August 2024–Present
Academic Resource Hub, Lafayette College Easton, PA
 - Lead two weekly collaborative review sessions to reinforce course material and support student learning.
 - Design worksheets, review activities, and exam-preparation materials.
 - Create a welcoming learning environment focused on conceptual understanding and peer-supported practice.

PROFESSIONAL SERVICE

- **Association for Computational Linguistics (ACL) Rolling Review** October 2025–November 2025
Secondary Reviewer with Dr. Sofia Serrano
 - Evaluated manuscript methodology, experimental design, and validity of scientific claims.
 - Drafted preliminary reviews refined through discussion with Dr. Serrano.
- **Machine Learning with Applications** April 2026–Present
Reviewer

EMPLOYMENT

- **Campus Tour Guide / Admissions Ambassador** September 2023–June 2025
Admissions Office, Lafayette College Easton, PA
 - Led campus tours for prospective students and families, presenting academic programs and student life through structured public presentations.
 - Represented Lafayette College to visitors and supported recruitment efforts through direct engagement with applicants.
- **IT Specialist Intern** June 2024–August 2024
WFMZ-TV Allentown, PA
 - Supported broadcast operations through installation, troubleshooting, and maintenance of newsroom and studio technology.
 - Assisted with teleprompter systems, workstation upkeep, and field deployment of technical equipment.

LEADERSHIP AND SERVICE

- **Lafayette DiscipleMakers Christian Fellowship** September 2023–Present
President; student leadership and audio/visual production
 - President, 2026–Present.
 - Selected to Oversight Team (2025) and Shepherding Team (2024), student leadership bodies responsible for organizational direction, decision-making, and ministry care.
 - Provide A/V production support for large-group gatherings.
- **Audio/Visual Specialist** July 2021–Present
Hope Alliance Church, Bethlehem, PA
 - Support weekly worship services through projection, livestream, and technical production.

ADDITIONAL INFORMATION

Languages: English (native); French (fluent, professional working proficiency)

Technical Skills: Python, Java, C; L^AT_EX; EnergyPlus; TensorFlow; PyTorch; data analysis and visualization workflows

Creative and Media Production: Trombone; jazz vocals; piano; composition; photography and photo editing; videography and video editing; live A/V production